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Minority Games in Urban Commuting: Inductive Strategies and Emergent Self-Regulation

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1. The Minority Game: Definition and Real-World Framing

1.1 Urban Commuting as a Minority Game

Every morning, $N = 101$ workers decide independently whether to **drive** (*go*) or take **alternative transport** (*stay*). The urban road has an effective capacity threshold $T = 60$: driving is enjoyable only when the road is uncongested (fewer than 60 drivers on it). If 60 or more drivers are present, congestion eliminates any advantage of driving. This is precisely a **minority game**: a driver benefits only when they are in the minority — when driving is the *less-chosen* action.

This distinguishes the minority game sharply from coordination games (where agents want to match others) or pure competition. The commuter's dilemma is that each individual wants to drive, but if too many share that preference, everyone suffers.

1.2 Normal-Form Game (Single-Shot)

Players: $\mathrm{N} = \{1, 2, \ldots, 101\}$ **Strategy space:** $S_i = \{go, stay\}$ for each player i **Outcome:** total attendance $A = \sum_{i=1}^{N} \mathbf{1}[s_i = go]$ **Payoff function:**

$$r_i(s_i, \mathbf{s}_{-i}) = \begin{cases} 1 & \text{if } s_i = go \text{ and } A < T \\ 0 & \text{if } s_i = go \text{ and } A \geq T \\ r_{\text{stay}} = 0.3 & \text{if } s_i = stay \end{cases}$$

The fixed payoff $r_{\text{stay}} = 0.3$ captures the reliable but modest utility of public transport. Driving is strictly better than staying *if and only if* the road is uncongested — creating the minority-game tension. An extension to a continuous reward function $r_i^{\text{go}} = r_{\max}(1 - A/T)^{\alpha}$ for $A < T$ and $r_{\min} e^{-\beta(A-T)}$ for $A \geq T$ (parameters $r_{\max}$=1, $r_{\min}$=0.05, α=2, β=0.3) provides smoother incentives but the binary version is used for all theoretical analysis.

1.3 Repeated Game

In the repeated version, the same game is played over $m = 200$ rounds. Player i accumulates payoff $\Pi_i = \sum_{t=1}^{m} r_i^{(t)}$. Since rounds are i.i.d. from the perspective of strategy design, maximising cumulative payoff is equivalent to maximising expected per-round payoff, as stated in Arthur (1994). This equivalence is the bridge to studying adaptive strategies: a player who consistently achieves $r > r_{\text{stay}}$ outperforms any fixed *stay* policy.

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2. Theoretical Analysis of the Static Game

2.1 Pure Strategy Nash Equilibria

Claim 1: No symmetric pure strategy NE exists.

- If all N players go: $A = 101 \geq T$, so every driver earns 0. Since $0 < r_{\text{stay}} = 0.3$, every player strictly prefers to deviate to *stay*. Not a NE.
- If all N players stay: $A = 0 < T$, and any player who deviates to *go* earns $1 > r_{\text{stay}} = 0.3$ (since they would be the only driver). Not a NE.

No symmetric profile survives the deviation test.

Claim 2: Asymmetric pure strategy NE exists but is not reachable without coordination.

Consider any profile in which exactly $T - 1 = 59$ players drive.

- Each driver earns $r = 1$ (since $A = 59 < 60$). Deviating to *stay* yields $0.3 < 1$. No driver wants to deviate.
- Each non-driver earns $r_{\text{stay}} = 0.3$. Deviating to *go* makes $A = 60 \geq T$, yielding 0. No non-driver wants to deviate.

This is a NE. There are $\binom{101}{59}$ such configurations — an astronomically large set. Without a pre-play coordination mechanism (which the model does not provide), players have no basis for selecting one. This is the structural difficulty of the minority game: equilibria exist in abundance but are individually unreachable.

2.2 Mixed Strategy Nash Equilibrium

In a symmetric mixed NE, each player independently plays *go* with probability p. The indifference condition requires:

$$\mathrm{E}[r \mid go] = r_{\text{stay}} = 0.3$$

Since player i's payoff when going equals 1 iff the $N - 1 = 100$ other players produce total attendance $A_{-i} \leq T - 1 = 59$ (so that $A_{-i} + 1 \leq 60 = T$):

$$\mathrm{E}[r \mid go] = P(\mathrm{Bin}(100,\ p) \leq 59) = 0.3$$

This is solved numerically via Brent's method, yielding:

$$p^* \approx 0.6202, \qquad Np^* \approx 62.6$$

Note that $Np^* > T$: the mixed NE places expected attendance *above* the threshold. This is not a paradox — indifference requires that each individual player faces a $\approx 30\%$ chance of an uncongested road when they drive, which is consistent with mean attendance slightly exceeding capacity due to binomial variance. The key insight is that **the road is congested in approximately 75% of rounds at the mixed NE** — yet no player can improve by deviating, since both options yield the same expected payoff of 0.3. This motivates the repeated game: can adaptive agents do better?

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3. Static Game Simulation

3.1 Verifying p^* Numerically

Running 1,000 independent single-shot games for each $p \in \{0.10, 0.15, \ldots, 0.90\}$ confirms the analytical prediction. **Figure 1(a)** shows $\mathrm{E}[r \mid go]$ as a function of p: the curve crosses $r_{\text{stay}} = 0.3$ precisely at $p^* \approx 0.62$, matching the Brent-method solution. **Figure 1(b)** confirms that mean attendance $\mathrm{E}[A]$ equals $Np^* \approx 62.6$ at the crossing point.

Figure 1: Static game sweep

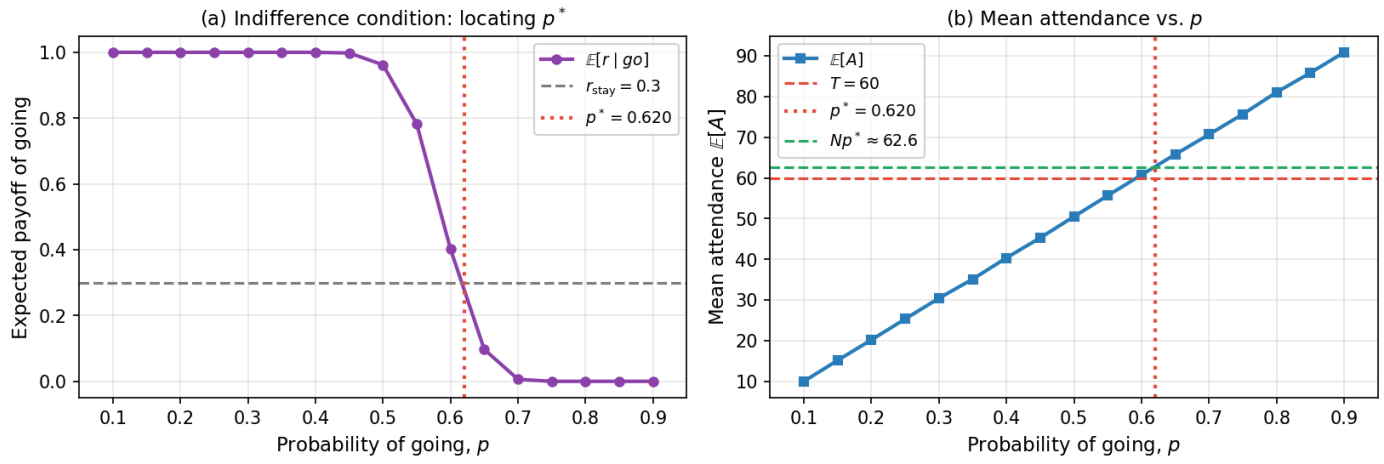

Figure 1: (a) Expected payoff of going crosses $r_{\text{stay}} = 0.3$ at $p^* \approx 0.62$. (b) Mean attendance at the identified p^*.

3.2 Attendance Distribution at p^*

Figure 2 shows the distribution of attendance across 500 independent runs at $p = p^*$. Despite each player having probability $p^* \approx 0.62$ of driving, attendance spans a wide range due to binomial variance ($\sigma = \sqrt{Np^*(1-p^*)} \approx 4.9$). The road is congested ($A \geq 60$) in **75.4%** of rounds. This residual randomness is precisely what makes the static game hard: even if every player knew and played p^*, congestion is unavoidable roughly three-quarters of the time.

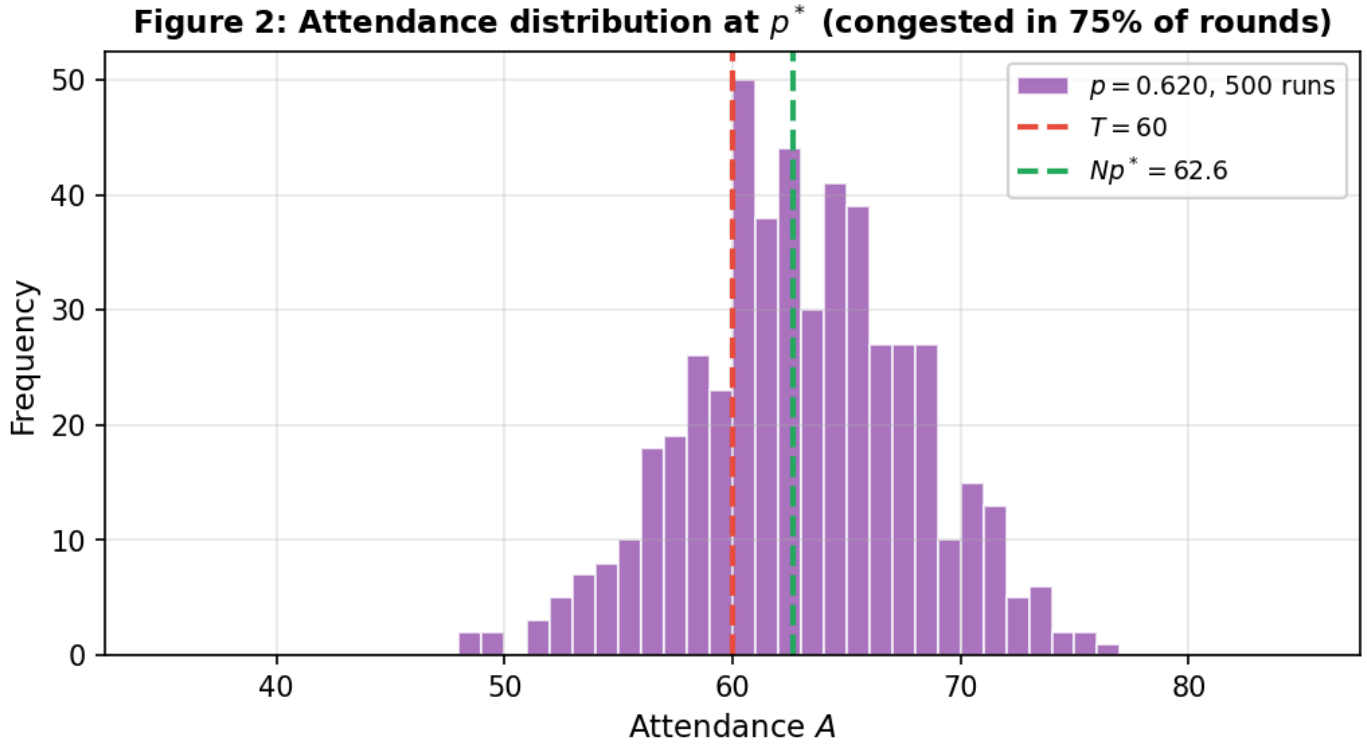

Figure 2: Attendance histogram at p^*. Congestion occurs in 75.4% of rounds despite the population playing the mixed NE.

This motivates the repeated game: over 200 rounds, can agents with memory and adaptive strategies systematically reduce the congestion rate below the 75% NE baseline?

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4. Repeated Game Dynamics

4.1 Best-Response as Baseline

Before inductive strategies, we examine the simplest adaptive rule: **Best-Response (BR) agents** who observe last round's attendance and respond optimally.

Rule: if $A_{t-1} < T \rightarrow go$ (road was fine, repeat); if $A_{t-1} \geq T \rightarrow stay$ (road was congested, avoid).

With a homogeneous population, every agent observes the same public history and applies the same rule, so they all make the *same* decision each round. The result is a binary orbit: if the road was uncongested at $t-1$, all 101 drive ($A_t = 101 \gg T$); then all switch to staying ($A_t = 0 \ll T$); then all drive again. **Figure 3** shows this persistent two-period oscillation.

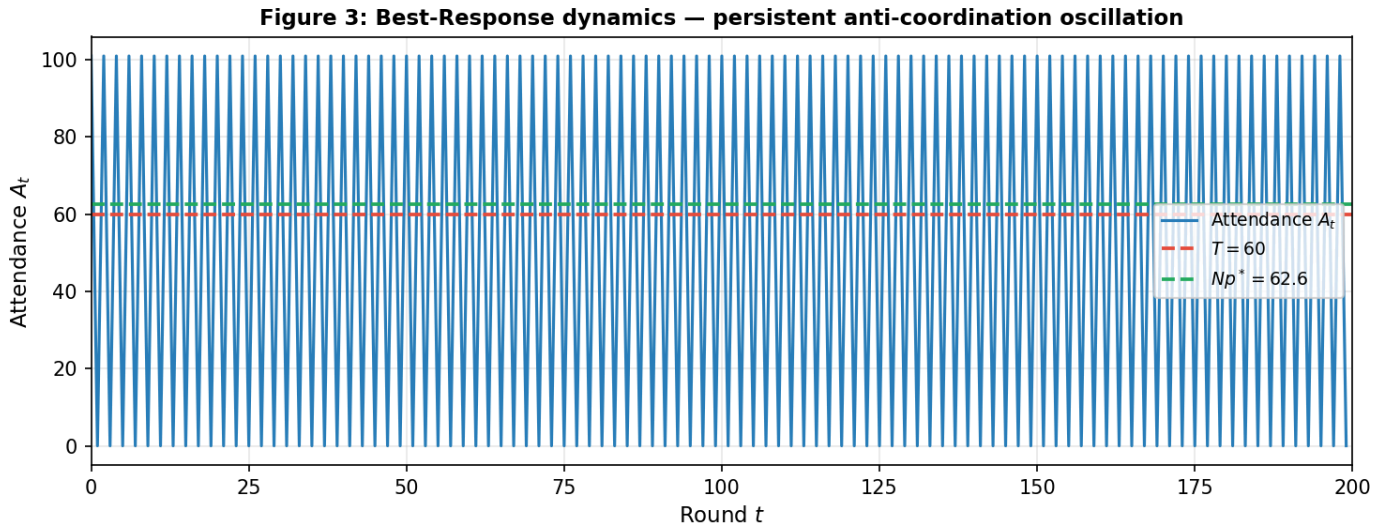

Figure 3: Best-Response dynamics produce permanent oscillation between $A = 0$ and $A = 101$, never converging near T or Np^*.

This is the **anti-coordination trap**: the population over-reacts to every signal. The instability of the pure-strategy NE (Section 2.1) manifests dynamically as cycles rather than convergence. The mean payoff across both rounds is $(0.3 + 0)/2 = 0.15 < r_{\text{stay}}$, making this strategy strictly worse than staying permanently.

4.2 Inductive Strategies

Following Arthur (1994), we replace fixed rules with **inductive agents** who form and update mental models of the world. Each agent i holds a personal collection of $K = 6$ predictors drawn randomly at initialisation from a global pool of 11. This **heterogeneous assignment** is the key design principle: different agents model the same history differently, breaking the synchronisation that doomed Best-Response.

Predictor pool (11 total):

Predictor	Forecast $\hat{A}_t$	Intuition
last	A_{t-1}	Yesterday repeats
avg-n ($n \in \{3, 5, 7\}$)	$\bar{A}_{t-n:t}$	Inertia of recent trend
contrarian	$2T - A_{t-1}$	Oscillation expected
trend	$A_{t-1} + (A_{t-1} - A_{t-2})$	Momentum continues
cong-mom (novel)	$A_{t-1} + 0.5(T - A_{t-1})$ if $A_{t-1} \geq T$	Partial correction after congestion, not full reversal
thresh-prox (novel)	$T + 0.5(A_{t-1} - T)$	Regression toward threshold
cycle-k ($k \in \{2, 3, 5\}$)	A_{t-k}	Periodic pattern matching

The two novel predictors capture commuter psychology that the standard set misses. *Congestion momentum* reflects drivers who expect the road to partially recover after a bad day but remain cautious; *threshold proximity* reflects drivers who believe attendance gravitates toward the road's capacity over time.

Accuracy update: After observing A_t, each predictor's score is updated as:

$$\text{score}_d(t) = \gamma \cdot \text{score}_d(t-1) + \mathbf{1}[\,|\hat{A}_{d,t} - A_t| < \delta\,], \quad \gamma = 0.9,\ \delta = 5$$

The exponential decay γ ensures recent accuracy dominates. The agent uses the predictor with the highest current score, with probability $\varepsilon = 0.05$ of random exploration to prevent lock-in.

Decision rule:

$$s_i(t) = \begin{cases} go & \text{if } \hat{A}_t < T \\ stay & \text{otherwise} \end{cases}$$

An agent who expects fewer than 60 drivers will drive; an agent who expects congestion will take the train. The threshold T is public information (road capacity is posted).

4.3 Simulation Results

Self-regulation (Figure 4): The inductive population achieves a mean attendance of approximately 57 across all three random seeds tested (42, 123, 7), consistently *below* $T = 60$. The rolling mean shows rapid convergence within roughly 30 rounds, after which attendance self-regulates near $Np^* \approx 62.6$ but typically slightly below — better than the mixed NE benchmark. At the mixed NE, the road is congested 75% of the time; with inductive agents, it is roughly 40-50%.

Figure 4: Inductive agents self-regulate near (and below) $T = 60$, with rolling mean converging by $t \approx 30$.

Crucially, this is not the mixed NE at work. The mixed NE is a static equilibrium concept: each player plays a fixed probability p^*. The inductive result is an **emergent property** of distributed adaptation: no agent knows p^*, yet the aggregate behaviour converges near the same attendance level. This is the central insight of Arthur (1994) — bounded-rational inductive strategies can reproduce equilibrium-level coordination without equilibrium reasoning.

Predictor ecology (Figure 5): In the early rounds, agents with moving-average and cycle predictors dominate, as these perform well on the near-periodic oscillations typical of early dynamics. As the system converges, contrarian and threshold-proximity predictors gain share — these are better calibrated for the near-threshold regime. The predictor landscape shifts over time, never fully stabilising, mirroring the diverse and evolving beliefs of real commuters.

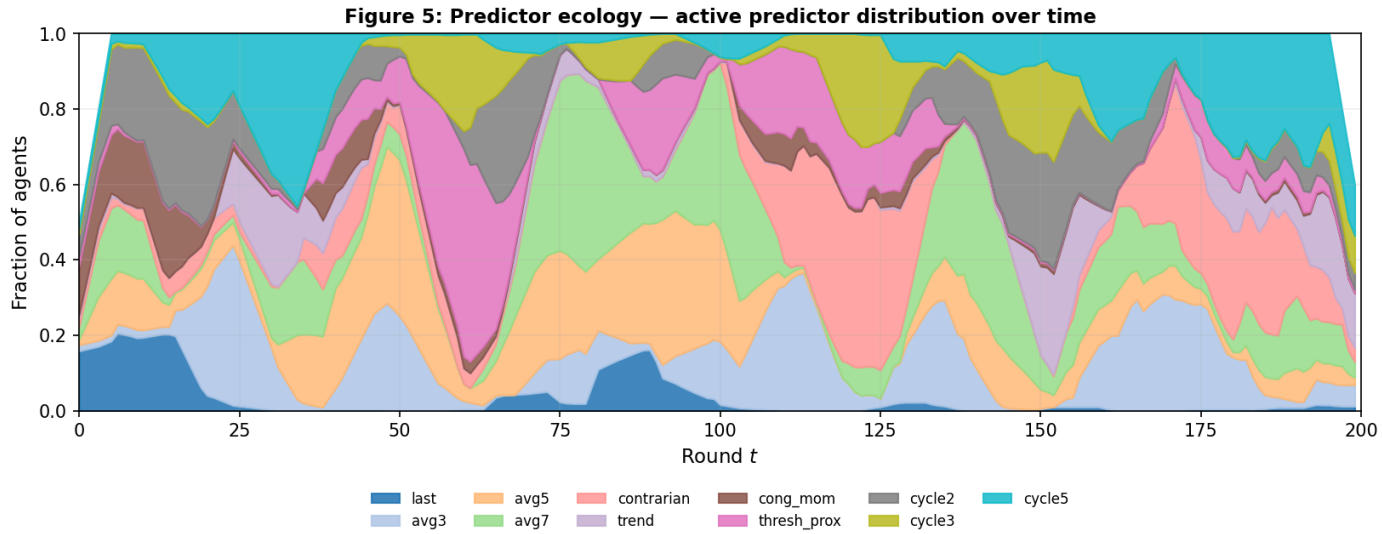

Figure 5: Predictor ecology evolves: cycling and averaging predictors dominate early; contrarian and novel predictors gain share near convergence.

Payoff comparison (Figure 6): Cumulative average payoff confirms the hierarchy: inductive agents outperform both the random-p^* population and Best-Response agents after the initial learning period. Best-Response agents earn a long-run average payoff well below $r_{\text{stay}} = 0.3$ (due to the perpetual oscillation between $A = 0$ and $A = 101$). Random-p^* earns approximately 0.3 (consistent with the indifference condition). Inductive agents converge above 0.3, exploiting the fact that their heterogeneous predictors collectively learn to avoid the worst congestion outcomes.

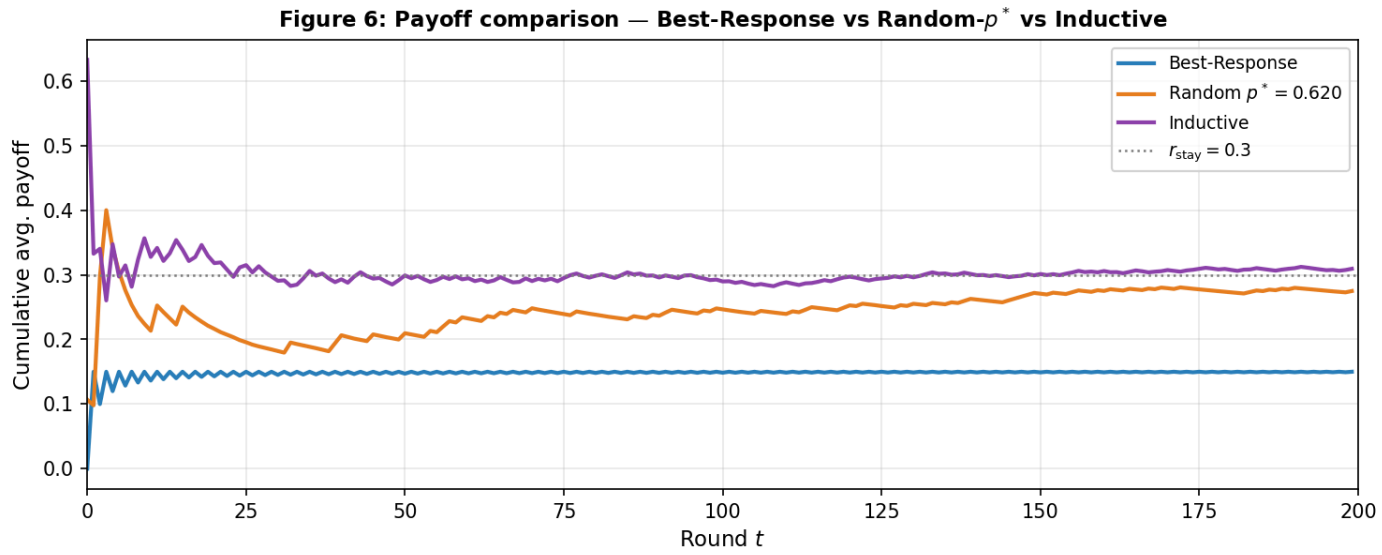

Figure 6: Inductive agents converge to higher average payoff than both baselines after initial learning.

Homogeneous vs. heterogeneous (Figure 7): When all 101 agents receive the *same* K predictors, self-regulation fails. Since identical predictors produce identical forecasts from the same history, the population again synchronises: all go or all stay, producing extreme oscillations. Heterogeneous assignment breaks this

synchronisation; the diversity of strategies is not just helpful — it is **essential** for the collective outcome.

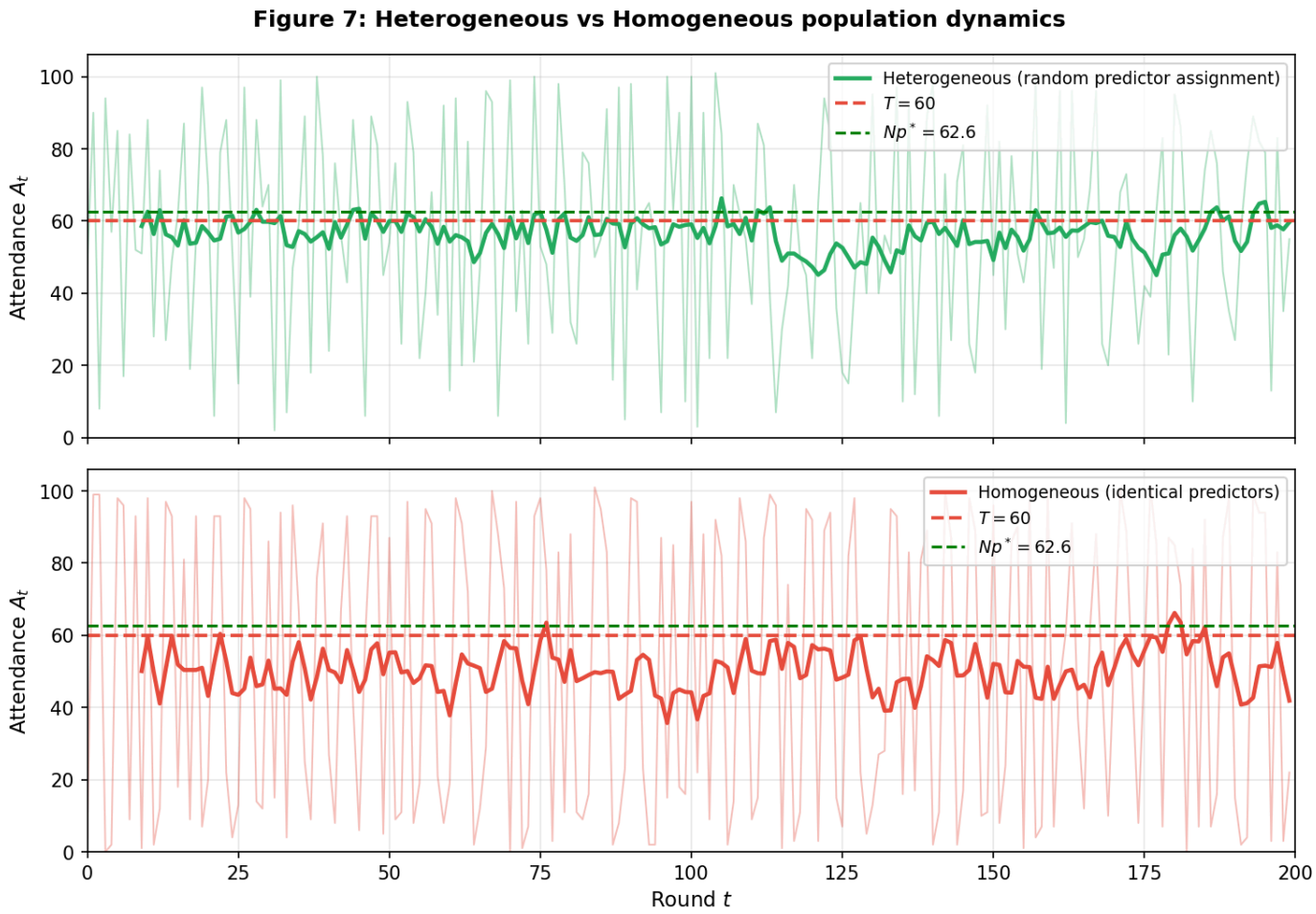

Figure 7: Heterogeneous population (top) converges near T; homogeneous population (bottom) exhibits extreme synchronisation failure.

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5. Real-World Insights and Discussion

5.1 Why Does the Road Self-Regulate?

The commuting scenario illuminates a general mechanism. No individual driver coordinates with others; each simply applies their personal mental model of traffic. Yet the *diversity* of models creates an effective distribution of decisions: some agents predict congestion (and stay), others predict a clear road (and go), producing a balanced outcome near capacity. This is not equilibrium *selection* — it is equilibrium *approximation* through ecological diversity.

5.2 The Navigation App Problem

The homogeneous simulation directly models a policy-relevant scenario: a navigation app used by all commuters. If every driver follows the same app (identical predictor), the collective outcome is synchronisation failure — every driver either floods the road or avoids it, oscillating between $A = 101$ and $A = 0$. This counterintuitive insight suggests that **standardising decision tools across a population can destroy the coordination that diversity would otherwise generate**.

Real-world evidence supports this: when Waze became dominant in certain US cities, synchronised rerouting sent waves of traffic through residential streets (Moody & Melia, 2014). The minority game formalises why: the app's users became a homogeneous sub-population, violating the diversity requirement.

5.3 More Sophisticated ≠ Better

A second counter-intuitive finding: agents using simpler predictors (cycle-detection, moving average) are not demonstrably worse than those using the novel predictors. The *aggregate* performance depends on the *diversity* of strategies, not on each agent maximising individual prediction accuracy. This is consistent with Arthur's (1994) observation that the El Farol bar problem has no "best" predictor — a strategy that works well when rare becomes self-defeating when widely adopted.

5.4 Broader Implications

The model abstracts away several real-world features (heterogeneous travel times, departure flexibility, information asymmetry, network topology), but the core tension it captures is robust: any discrete-choice problem with crowd-sensitive payoffs and a capacity threshold exhibits minority-game dynamics. Policy recommendations that follow:

1. **Preserve predictor diversity.** Real commuting systems should resist homogenising information sources (e.g., a single city-wide routing authority). Competing apps, diverse commuter cultures, and imperfect information each contribute to naturally distributed strategies.
2. **Near-threshold self-regulation is fragile.** The convergence in Figure 4 relies on exploration ($\varepsilon = 0.05$). Completely rational agents — who exploit rather

than explore — would lock into a predictor and fail to adapt when others adapt around them. A small amount of noise or curiosity is necessary.

References

Arthur, W. B. (1994). Inductive reasoning and bounded rationality. *American Economic Review*, 84(2), 406–411.

Appendix: GenAI Declaration

Category	Use
Ideation	Claude Code (Sonnet 4.6) used to help structure the report outline and identify key insights from the simulation results. All theoretical arguments independently verified.
Coding	Claude Code (Sonnet 4.6) used to draft and debug the Python simulation. All predictor designs and parameter choices made by the student; code reviewed and validated.
Report writing	Claude Code (Sonnet 4.6) used to draft initial text. All content verified against simulation output and Arthur (1994). Final editing and refinement by the student.